

[LinkX Ideas] The AGI inflection point has not yet arrived, and technology priority remains the top priority

Abstract: The Intelligence Ceiling of Large Models, Agent Evolution, and Application Implementation Paths



Editor's Note:

At a time when there is no clear inflection point for AGI yet, how to identify the main line of technological evolution and the stage-specific opportunities is a common concern for both the industry and the investment community. Based on a series of recent industry exchanges and technical discussions organized by Xinglian Capital, this article systematically sorts out the judgments related to the AGI path, the evolution of large model capabilities, and the implementation of applications.

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The inflection point of AGI has not yet arrived, and the industry remains in the technology-prioritized stage rather than rapidly shifting towards product orientation.

In recent closed-door discussions organized by Starlink Capital, many guests from the industry and academia have formed a unanimous judgment on the implementation path and development stage of AGI.

At a stage where current technological capabilities are still in a phase of linear growth, on the one hand, continuously exploring the upper limit of model intelligence remains the most important task; on the other hand, how to effectively activate and unleash the capabilities that models already possess in real, complex scenarios has also become an unavoidable real-world challenge. These two points constitute the core issues currently attracting industry attention.

Based on the consensus and divergence in the closed-door discussions, this paper systematically examines the key development priorities, practical constraints, and potential evolutionary directions of the next stage in the AI field from multiple dimensions, including the enhancement of base model capabilities, the activation of alignment and inference capabilities, the evolution of Agent forms, and the key technical challenges faced during the process of models moving into the real world.

Inflection Point Not Yet Reached: Exploring the Upper Bound of Intelligence Remains a Priority, and Scaling Remains the Most Efficient Approach

Pre-training enables large models to have already grasped common-sense knowledge of the world and possess simple reasoning abilities. More data, larger parameters, and more saturated computation remain the most efficient ways to scale the base model.

Over the past two years, the industry has made numerous attempts in areas such as MultiModal Machine Learning and end-to-end speech, but practice has proven that what truly determines the upper limit of model intelligence remains text. The knowledge volume and basic reasoning ability of current models both rely on "extensive reading," that is, text data that carries human

common-sense knowledge. From the perspective of first principles, to explore the upper limit of intelligence, the first core element is larger-scale data, especially common-sense knowledge data (real data, synthetic data, and distilled data).

Meanwhile, although everyone is exploring more efficient ways at both the model layer and infra (infrastructure) layer, scale will still be the most effective path to enhancing model capabilities in the foreseeable future—besides larger-scale data, larger model parameter sizes, and more computing power investment remain necessary.

Key to Unleashing the Capabilities of Existing Models: Activation Alignment and Enhanced Reasoning Ability

Activating alignment and enhancing inference capabilities, especially activating more comprehensive long-tail capabilities, is another key to ensuring the effectiveness of the model.

Even if a model has common sense knowledge, its capabilities are difficult to unleash if it cannot be effectively activated. In the past, alignment mainly relied on methods such as SFT and CoT, while currently, reinforcement learning is gradually becoming the mainstream approach: by constructing scenarios and data, the model self-optimizes in reinforcement learning to activate the upper bound of intelligence.

However, the real challenge lies in that the emergence of general benchmarks, on the one hand, can evaluate the general effectiveness of models, but on the other hand, may also lead to overfitting in many models. Meanwhile, the number of evaluation benchmarks is limited, while the real-world application scenarios are almost infinite. With the emergence of a large number of agents, the complexity of alignment has increased significantly—if model alignment in the past was more like testing knowledge points such as Chinese and mathematics, then agents are closer to real work environments, with diverse job types, complex situations, and even requiring a certain level of "emotional intelligence". For example, booking flights and hotels essentially involves aligning with human behavior by introducing Agent data, thereby activating the model's capabilities. Therefore, the breakthrough to be achieved in the next stage is to enable the model to align more quickly and effectively with long-tail real-world scenarios. Mid-training and Post-training will make it possible to achieve rapid alignment of more scenarios and strong reasoning capabilities.

Agent Evolution: Cross-Environment Generalization and Transfer Remain Difficult Problems, and the Simple Solution is to Construct Environmental Data + Reinforcement Learning

Agent is a milestone in the expansion of model capabilities and also a key indicator of AI models entering the real (virtual/physical) world of humans. Without Agent capabilities, large models will remain in the (theoretical learning) stage and struggle to convert knowledge accumulation into real productivity.

In the coming year, a large number of Agents will continue to emerge - this prediction is no longer new. But to promote the large-scale application of Agents, the core lies first in understanding that the essence of AI is not to create entirely new capabilities, but rather **to systematically replace some of human work or capabilities**. This is also the key factor determining the inevitable explosion of Agents.

Secondly, although early Agents were mainly implemented through model applications, current models can now directly integrate Agent data into the training process, thereby enhancing the model's generality. However, the challenge lies in generalization and transfer between different Agent environments. Therefore, the simplest path at the application layer, as previously mentioned, is to continuously introduce data from more different Agent environments and conduct reinforcement learning for different environments.

Key Points of Model Capability Breakthrough: Memory, Online Learning, and Self-Assessment

Achieving model memory is an essential capability for models to be applied in real-world environments and also a problem that must be addressed currently.

An analogy is made with the human memory structure: humans have short-term memory (prefrontal cortex), medium-term memory (hippocampus), long-term memory (distributed cerebral cortex), and external memory systems such as human history (e.g., historical records, Lingo). For large models, context, RAG, and model parameters may respectively correspond to different memory stages in humans, but how to truly implement these capabilities remains a key challenge. A feasible approach is to compress memory and simply store it in the context. If the model can support a sufficiently long context length, it is theoretically possible to achieve

short-, medium-, and long-term memory. However, how to iterate model knowledge and how to update model parameters are challenges that need to be overcome.

Meanwhile, on this basis, online learning has also become a key issue that cannot be bypassed. Most current large models adopt the method of periodic retraining. This leads to a series of problems: on the one hand, the model cannot truly achieve self-iteration, but self-iteration must be an essential ability for the model in the next stage; on the other hand, retraining is relatively resource-intensive and will discard a lot of interactive data during the process. To achieve online learning, a key point is the model's self-assessment - the model must first know whether it is right or wrong, even if it only knows the right or wrong in a probabilistic sense, the model can then identify the optimization target and achieve self-improvement. Therefore, in the next stage, the challenge to be addressed is to build a model self-assessment mechanism. Such mechanisms may also give rise to new scaling paradigms, perhaps leading to the emergence of capabilities driven by continual learning, real-time learning, or online learning.

Application Layer: Build applications around different job types and keep them thin enough to adapt to the evolution of model capabilities

Regarding the application layer, there are two main judgments: First, in the context of the increasingly end-to-end development of large models, model R&D and model application inevitably need to be combined. The application layer must be made sufficiently "thin" to quickly adapt to the changes in the capabilities of the base model, and rapidly launch, iterate, and validate value; Second, building model applications should start from the perspective of helping people and creating new value. If an AI software is not used by anyone and thus fails to create value, it is destined to lack vitality.

Specifically, what exactly should AI software do? Returning to first principles, AI does not need to create new applications; its essence is to simulate, replace, or assist humans in achieving certain tasks that humans must perform (certain types of jobs). From this perspective, future applications will mainly fall into two categories: AI-ifying past software, where the parts that originally required human participation are replaced by AI; and creating AI software that aligns with a particular public need to replace human work.

Embodied Intelligence: Unlocking Model Capabilities in More Real-World Scenarios Requires Solving Data Scale and Hardware Technology Issues

Outside the virtual world, corresponding to the emergence of Agents, is the development of embodied intelligence. Currently, robot brains already possess strong intelligence, and the key to the next step lies in aligning and activating the capabilities of large models with robot behavior. As hardware technology gradually matures, embodied intelligence is expected to unleash model capabilities in more real-world scenarios. However, just like the problem of poor generalization faced by Agents, it is basically impossible to activate general embodied capabilities through few-shot learning. This requires breaking through the problem of large-scale data acquisition.

MultiModal Machine Learning: A long-term important direction, and the efficient approach is to explore text, multi-modal understanding, and multi-modal generation in parallel

Although MultiModal Machine Learning has not yet provided substantial assistance in exploring the upper limit of AGI's intelligence, it is undoubtedly an important direction. Currently, the effective path remains the parallel development of text, multi-modal understanding, and multi-modal generation, along with appropriate exploration of the combination of the three to seek the emergence of new capabilities. Of course, exploring the combination of the three not only requires courage in technological exploration but also long-term and continuous capital investment.

Domain Model: Rationally Viewing Phased Opportunities

Under the AGI endgame scenario, domain models are likely to be overwritten by base models, and domain data, processes, and Agent data will gradually be incorporated into the main model. However, during the stage when AGI has not yet been achieved, as long as there is a determination for continuous investment, and with the possession of domain data and close coupling with the base model, there is still an opportunity to establish a leading edge within a specific competitive window.